BKP: An R Package for Beta Kernel Process Modeling

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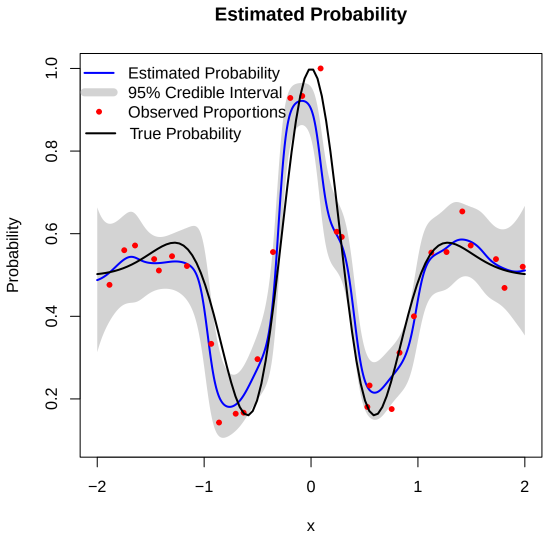
Contents

Estimating an Unknown Probability Surface



- **Observation model**

$$y_i \mid \pi(\boldsymbol{x}_i) \sim \mathrm{Binomial}(m_i, \pi(\boldsymbol{x}_i)) .$$

- **Objective**

  Estimate the underlying probability surface $\pi(\boldsymbol{x})$ and quantify its pointwise uncertainty.

Existing Approaches and the Remaining Gap

Logistic regression

$$\pi(\boldsymbol{x}) = \operatorname{logit}^{-1}\left(\boldsymbol{x}^\top \boldsymbol{\beta}\right)$$

Fast and interpretable

Limited nonlinear flexibility

Gaussian process classification

$$\pi(\boldsymbol{x}) = \operatorname{logit}^{-1}\{f(\boldsymbol{x})\}, \quad f(\boldsymbol{x}) \sim \mathrm{GP}$$

Flexible probability surfaces

Approximate posterior inference

Motivation

Can we retain nonlinear flexibility while performing direct, closed-form updating on the probability scale?

Contents

Beta Kernel Process: Direct Probability-Scale Updating

Beta prior

$$\pi(\boldsymbol{x}) \sim \text{Beta}\left\{\alpha_0(\boldsymbol{x}), \beta_0(\boldsymbol{x})\right\}.$$

Prior information on the probability scale

$+$

Local likelihood

$$k_i(\boldsymbol{x}) = k(\boldsymbol{x}, \boldsymbol{x}_i; \boldsymbol{\theta}),$$

$$S(\boldsymbol{x}) = \sum_{i=1}^{n} k_i(\boldsymbol{x}) y_i,$$

$$F(\boldsymbol{x}) = \sum_{i=1}^{n} k_i(\boldsymbol{x})(m_i - y_i).$$

Nearby observations receive greater weight

$\Rightarrow$

Beta posterior

$$\pi(\boldsymbol{x}) \mid \mathcal{D} \sim \text{Beta}\left\{\alpha_n(\boldsymbol{x}), \beta_n(\boldsymbol{x})\right\},$$

$$\alpha_n(\boldsymbol{x}) = \alpha_0(\boldsymbol{x}) + S(\boldsymbol{x}),$$

$$\beta_n(\boldsymbol{x}) = \beta_0(\boldsymbol{x}) + F(\boldsymbol{x}).$$

Closed-form posterior inference

Key idea

Local kernel weighting $+$ **Beta–binomial conjugacy**

$\Longrightarrow$ **flexible modeling with closed-form posterior inference.**

Closed-Form Posterior Summaries

Probability estimation

$$\widehat{\pi}(\boldsymbol{x}) = \frac{\alpha_n(\boldsymbol{x})}{\alpha_n(\boldsymbol{x}) + \beta_n(\boldsymbol{x})}$$

Posterior mean of the unknown probability surface

Pointwise uncertainty

$$\mathrm{Var}\{\pi(\boldsymbol{x}) \mid \mathcal{D}\} = \frac{\widehat{\pi}(\boldsymbol{x})\{1 - \widehat{\pi}(\boldsymbol{x})\}}{\alpha_n(\boldsymbol{x}) + \beta_n(\boldsymbol{x}) + 1}$$

$$\mathrm{CI}_{1-\delta}(\boldsymbol{x}) = \left[Q_{\delta/2}(\boldsymbol{x}), Q_{1-\delta/2}(\boldsymbol{x})\right]$$

Credible intervals follow from Beta posterior quantiles

Count prediction

$$Y_{\mathrm{new}} \mid \mathcal{D}, \boldsymbol{x}, m_{\mathrm{new}} \sim \text{Beta-Binomial}\left\{m_{\mathrm{new}}, \alpha_n(\boldsymbol{x}), \beta_n(\boldsymbol{x})\right\}.$$

Closed-form prediction for future binomial counts

Computational implication

Conditional on the kernel hyperparameters, posterior summaries and predictive distributions are available without numerical posterior approximation.

Prior Specification

Non-informative prior

$$\pi(\boldsymbol{x}) \sim \mathrm{Beta}(1,1).$$

Default uniform prior when no external information is available

Informative prior with fixed mean

$$\alpha_0(\boldsymbol{x}) = r_0 p_0, \quad \beta_0(\boldsymbol{x}) = r_0(1-p_0)$$

$$\mathbb{E}\{\pi(\boldsymbol{x})\} = p_0, \quad \alpha_0(\boldsymbol{x}) + \beta_0(\boldsymbol{x}) = r_0$$

A fixed prior mean p_0 with precision controlled by r_0

Data-adaptive prior

$$p(\boldsymbol{x}) = \sum_{i=1}^{n} \frac{k(\boldsymbol{x},\boldsymbol{x}_i)}{\sum_{i=1}^{n} k(\boldsymbol{x},\boldsymbol{x}_i)} \frac{y_i}{m_i}, \quad r(\boldsymbol{x}) = r_0 \sum_{i=1}^{n} k(\boldsymbol{x},\boldsymbol{x}_i)$$

$$\alpha_0(\boldsymbol{x}) = r(\boldsymbol{x})p(\boldsymbol{x}), \quad \beta_0(\boldsymbol{x}) = r(\boldsymbol{x})\{1-p(\boldsymbol{x})\}.$$

The prior mean follows the local empirical proportion, while the prior precision adapts to the local kernel mass

Scaled Distance and Length-Scale Parameters

Scaled distance

$$h(\boldsymbol{x}, \boldsymbol{x}'; \boldsymbol{\theta}) = \sqrt{\sum_{j=1}^{d} \left(\frac{x_j - x'_j}{\theta_j} \right)^2}.$$

- Smaller θ_j: more localized borrowing and greater sensitivity along dimension j
- Larger θ_j: broader borrowing over longer distances

Length-scale structure

Isotropic: $\theta_1 = \cdots = \theta_d = \theta$

Anisotropic: $\boldsymbol{\theta} = (\theta_1, \ldots, \theta_d)$

An isotropic kernel uses a common length scale across dimensions, whereas an anisotropic kernel allows dimension-specific borrowing ranges

Modeling role

The length-scale parameters determine how rapidly information borrowing decreases with distance.

Kernel Functions

Implemented kernels

$$\text{Gaussian:} \quad k(h) = \exp(-h^2),$$

$$\text{Matérn } 5/2: \quad k(h) = \left(1 + \sqrt{5}h + \frac{5}{3}h^2\right)\exp(-\sqrt{5}h),$$

$$\text{Matérn } 3/2: \quad k(h) = \left(1 + \sqrt{3}h\right)\exp(-\sqrt{3}h),$$

$$\text{Wendland:} \quad k(h) = (\zeta h + 1)\max\{0, 1-h\}^{\zeta}, \quad \zeta = \lfloor d/2 \rfloor + 3.$$

Gaussian and Matérn

Noncompact support

Every observation at a finite distance receives a positive weight, although the weight decreases with distance.

Wendland

Compact support

Observations with $h \geq 1$ receive exactly zero weight.

LOOCV-Based Length-Scale Selection

For each candidate $\boldsymbol{\theta}$, evaluate predictive performance by leave-one-out cross-validation.

Leave-one-out prediction

$$\widehat{\pi}^{-i}(\boldsymbol{x}_i; \boldsymbol{\theta})$$

denotes the predicted probability at $\boldsymbol{x}_i$ after excluding the ith observation.

All leave-one-out predictions are computed from a single kernel matrix by setting its diagonal entries to zero.

Brier score

$$\mathrm{BS}(\boldsymbol{\theta}) = \frac{1}{n} \sum_{i=1}^{n} \left\{ \widehat{\pi}^{-i}(\boldsymbol{x}_i; \boldsymbol{\theta}) - \widetilde{\pi}_i \right\}^2$$

Emphasizes squared probability-estimation error

Log-loss

$$\mathrm{LL}(\boldsymbol{\theta}) = -\frac{1}{n} \sum_{i=1}^{n} \Big[\widetilde{\pi}_i \log \widehat{\pi}^{-i}(\boldsymbol{x}_i; \boldsymbol{\theta}) + (1 - \widetilde{\pi}_i) \log\{1 - \widehat{\pi}^{-i}(\boldsymbol{x}_i$$

Penalizes overconfident probability errors more strongly

Multi-Start Hyperparameter Optimization

Log-scale reparameterization

$$\gamma_j = \log_{10}(\theta_j), \qquad \theta_j = 10^{\gamma_j}.$$

$$d_\gamma = \begin{cases} 1, & \text{isotropic,} \\ d, & \text{anisotropic.} \end{cases}$$

The transformation enforces positive length-scale parameters

Optimization workflow

1. Generate $n_0 = 10d_\gamma$ initial values using a Latin hypercube design
2. Run the derivative-free SBPLX algorithm from each initial value
3. Retain the solution with the smallest LOOCV loss

$$\widehat{\boldsymbol{\gamma}} = \underset{\boldsymbol{\gamma} \in [-3,3]^{d_\gamma}}{\operatorname{argmin}} \; L(\boldsymbol{\gamma}; \mathcal{D})$$

Why multi-start?

The LOOCV objective may be non-convex, so multiple initial values reduce sensitivity to a single local optimum.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THANK YOU